

Fig. 5: Actual-size PCB

12V relay. The 4049 input threshold and CD4050 switching threshold vary with supply voltage, temperature, IC manufacturer, and device characteristics.

Construction and testing

The actual-size, single-sided PCB layout for the on-delay timer is shown in Fig. 5, and the component layout in Fig. 6. After assembling the circuit, mount it in a suitable enclosure. Fix CON1 and CON2 on the rear panel, and switches and LEDs on the front panel. Mount potentiometer VR1 on the front panel for convenient delay adjustment. Fig. 7 is an alternative modified off-delay timer, so its PCB layout is not provided.

Alternatively, the circuit can be assembled on a Veroboard or general-purpose PCB. A 12V DC SMPS module is recommended for powering the circuit, although a 12V transformer with a rectifier and filter can also be used.

In the on-delay circuit, LED1 glows immediately after power-on, while LED2 glows after the preset delay when the relay energises. In the off-delay circuit, both LEDs initially glow; after the preset delay, the relay de-energises and LED2 goes off. After successful testing, connect the appliance through the relay contacts and a suitable AC outlet.

PARTS LIST (FIG. 3 ON-DELAY TIMER)

Semiconductors:	
IC1 (IC1A-IC1G)	- CD4050 hex buffer
T1	- 2N3904 NPN transistor
D1, D2	- 1N4007 rectifier diodes
LED1, LED2	- 5mm red and green LEDs
Resistors (all 1/4-watt, ±5% carbon):	
R1, R4	- 4.7kΩ
R2, R3	- 1kΩ
VR1	- 1MΩ potmeter
Capacitors:	
C1	- 470µF, 25V electrolytic
C2	- 220µF, 16V electrolytic
Miscellaneous:	
RL1	- 12V SPDT relay
CON1, CON2	- 2-pin connectors
BATT.1	- 12V battery/12V SMPS supply
LOAD	- Anything like refrigerator or an AC
S1	- Push-to-on switch

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